ZUL IKRAM MUSADDIK RAYAT

Lalbagh Road, Dhaka • +880 1580-863471 • rayathossain49@gmail.com • linkedin.com/in/zim-rayat
• github.com/devrayat000 • v2.rayat.dev • ORCID: 0009-0003-3661-7731

Summary

M.Sc. in Computing student at BUET CSE and Mechanical Engineering graduate from BUET intentionally transitioning into Software Engineering, bringing strong mathematical foundations, systems modeling, and production experience. Completed a 5-month software engineering internship at RubizCode IT developing backend APIs and AI systems in NestJS, FastAPI, and Redis for 100,000+ users. Comfortable across the stack, from distributed backend pipelines and on-device runtimes to web platforms. Evening/flexible graduate schedule allows 100% full-time professional availability for software engineering roles.

Education

M.Sc. in Computing **2026 – Present**
Department of Computer Science and Engineering, BUET
- Evening / flexible program • Full-time professional software engineering availability
- Focus: Advanced Algorithms, Distributed Systems, Software Architecture

B.Sc. in Mechanical Engineering **June 2026**
Bangladesh University of Engineering and Technology (BUET)
- CGPA: 3.10 / 4.00 • Former Class Representative

Higher Secondary Certificate (HSC) **2020**
Notre Dame College, Dhaka
- GPA: 5.00 / 5.00

Software Engineering Experience

Software Engineering Intern - Backend & AI Systems **May 2025 – Sept 2025**
RubizCode IT
- Built and scaled **Taalaash**, an academic search engine serving **100k+ users**, **30k+ active students**, and **2k+ daily queries** with **99.9% uptime**.
- Engineered an OCR & text-extraction pipeline using **FastAPI** and fine-tuned **Tesseract OCR**, indexing **20k+ textbook pages** with **98% character recognition accuracy** for Bangla script.
- Implemented semantic search, full-text indexing, and vector reranking using **PostgreSQL**, **Algolia**, and **Pinecone**, lifting search relevance by **35%** and cutting page indexing time by **80%**.
- Developed **TopperOn Doubts AI**, architecting an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline with **FastAPI**, **LangChain**, and **OpenAI API**, reducing query latency by **35%** and boosting context-relevance scores by **2 points**.
- Architected modular backend **microservices** using **NestJS**, **Prisma**, and **PostgreSQL**, incorporating **Redis** caching and multi-layer CDN strategies to improve data throughput and system efficiency by up to **95%**.
- Implemented real-time interactive exam and polling engines using **WebSockets** and asynchronous message queues to handle concurrent live student participation.
- Containerized microservices using **Docker** and established deployment workflows on **GCP** managed behind **Traefik** reverse proxy and **Cloudflare** for automated routing, SSL, and load balancing.

Extracurricular Activities

President **Mar 2025 – Present**
BUET Animata - Animation & VFX Club
- Served as Founding Assistant General Secretary; organized workshops on basic animation and 3D modeling.
- Designed and animated the official club logo and visual assets.

Web Master **Oct 2022 – Oct 2024**
Team Interplanetar - BUET Mars Rover Team
- Developed and maintained the home website of Team Interplanetar containing detailed descriptions of its products, findings and achievements.
- Forged a new certifications system in the website to provide certifications on engineering aspects of mars rovers.

Skills

Languages: JavaScript, TypeScript, Python, Go, Rust, C/C++, C#, SQL
Frontend & Mobile: React, Next.js, React Native, Expo, Redux, Svelte, Tailwind CSS, Framer Motion, React-Query, Jotai, Zustand, Vite
Backend: Node.js, Nest.js, Express.js, Hono, FastAPI, Django, .NET Core, GraphQL, REST
AI & Machine Learning: RAG (Retrieval-Augmented Generation), LangChain, Pinecone, OpenAI API, Tesseract OCR, TensorFlow, Scikit-Learn, NumPy, Pandas
Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis (Streams, Pub/Sub), Cloud Firestore, Pinecone (Vector DB)
DevOps & Infrastructure: Docker, Docker Swarm, Traefik, Cloudflare, GitHub Actions, AWS, Google Cloud Platform (GCP), Vercel, Heroku
Tools & Middleware: RabbitMQ, Celery, FFmpeg, Git, Linux, Webhooks, gRPC, LaTeX

Notable Projects

TrueNews - AI News Verification Platform **Personal**
TypeScript, Node.js, AI Pipeline, Motia, Vite
- Built an AI-powered backend for real-time media analysis, source-credibility scoring, and automated fact-checking.
- Live at truenews.wasmer.app

AIVIBLE - AI-Visibility SaaS Platform **Personal**
Next.js, AI, LLM, SaaS
- Shipped an AI-visibility SaaS for the travel industry, pairing an LLM-powered itinerary planner with tools that help local businesses surface correctly in AI chatbot responses (e.g., ChatGPT, Gemini).
- Live at ai-audit-seven.vercel.app

ScabieSense - AI Health Screening App **BUET IAT Research**
React Native, Expo, TypeScript, Deep Learning
- Built a cross-platform mobile app for early scabies detection using a custom deep learning model, developed for BUET's Cyber Physics Intelligence Lab and field-tested on university dorm students.
- Source code: devrayat000/scabies-net

Video Transcoding Pipeline **Personal**
Go, Redis Streams, FFmpeg, React/Vite
- Designed a multi-service video transcoding system with **Redis Streams** consumer groups for durable, at-least-once job distribution across horizontally-scalable workers.
- Built the worker backend in **Go** for high-throughput **FFmpeg** processing, paired with a real-time **React/Vite** dashboard for monitoring job status and failure modes.
- Source code: devrayat000/video-processing

Retina AI - Distributed Document Ingestion Pipeline **Freelance**
TypeScript, Python, Redis (BullMQ), S3, Docker, Gemini, AI SDK
- Designed a multi-service, queue-based pipeline: a Node.js API enqueues jobs to Redis (BullMQ); horizontally-scalable, independently-deployed Python workers pull jobs, extract and OCR documents (PyMuPDF, Tesseract), and hand off structured output to a downstream embedding queue via S3-compatible object storage.
- Decoupled extraction and embedding into stateless services that scale independently (configurable worker replica count) and communicate only through queues and object storage, with Docker Compose and PM2 deployment paths.
- Source code *available on request*

Focus Stream - On-Device AI Focus Journal **Personal**
Rust, Tauri 2, Llama 3.2 (on-device), SQLite
- Built a fully offline desktop app running a local Llama 3.2 model (via mistral.rs) for on-device AI-generated daily summaries, with no cloud dependency or telemetry.
- Designed a compile-time-selectable inference backend (in-process linking vs. a dynamically-loaded DLL) to support fast dev iteration and a slim production binary across x64/arm64 builds; published to the Microsoft Store.
- Source code *available on request*

Assistive BdSL Gesture Generation on a Robotic Hand **Undergraduate Project**
Python, MediaPipe, TensorFlow, Arduino, Flex Sensor Servo Motor
- Built a real-time speech-to-sign pipeline (ASR → landmark-based MLP classifier → servo control), achieving 91.11% test accuracy across 38 gesture classes on a 3D-printed robotic hand prototype.

Rover Camera Feed **Team Interplanetar - BUET Mars Rover Team**
Python, OpenCV, Flask, WebSockets, WebRTC, Live-Stream
- Built a high-performance multi-camera streaming system for robotics teleoperation, using low-latency WebSocket transmission and OpenCV-based video processing across simultaneous camera feeds.
- Source code: devrayat000/rover_camera_feed

Problem Solving, Certifications & Publications

- **Algorithmic Problem Solving:** Active C++ practice on **LeetCode** and **NeetCode** (devrayat000/neetcode-submissions).
- **Published IEEE Papers (3):** Author/co-author on 3 peer-reviewed IEEE publications: Hard-Constraint Navier-Stokes PINNs, Spatiotemporal PM2.5 Attention Forecasting, and 6-DOF Robotic Hand BdSL Kinematics.
- **Foundational CS Certifications:** CS50x (Harvard) • Algorithms Specialization (Stanford) • Algorithms, Part I & II (Princeton).
- **Honors & Awards:** **Finalist**, Build4Democracy Hackathon (UNDP, 2025) • **Finalist**, Innovation Founder (RISE UIHP, 2024) • **1st Place**, Speak Out For Engineers (IMechE BUET Student Chapter, 2023) • **Technical Scholarship**, BUET (2022, 2023).

Reference

Dr. Mohammad Mamun
Professor
Dept. of Mechanical Engineering, BUET
Email: dmmamun@me.buet.ac.bd

Dr. Mohammad Jane Alam Khan
Associate Professor
Dept. of Mechanical Engineering, BUET
Email: ronin@me.buet.ac.bd